

Letter of Inquiry

Journal of Quantitative Description: Digital Media

September 15, 2023

To study the diffusion or discussion of misinformation on social media, computational researchers commonly use stories that were factchecked by experts (Vosoughi et al., 2018), or track blacklisted domains (González-Bailón et al., 2023; Guess et al., 2020; Grinberg et al., 2019). A prominent domain list is curated by the organization NewsGuard (Robertson et al., 2023; Lasser et al., 2022), however, it is proprietary and so far only validated for the U.S. context by Lin et al. (2023). Therefore, this proposal outlines a descriptive review of the NewsGuard list over time and across countries, and is guided by the question: How reliable is a domain-based list, such as NewsGuard, to measure misinformation online?

What is being described?

NewsGuard applies an expert-based approach; analysts assign and regularly update overall reliability ratings (0 to 100) based on nine binary criteria (see here). For the U.S. context, the list has shown high agreement with other expert-rated lists (Lin et al., 2023), but it is crucial to assess the usefulness for research beyond the English-speaking world. Moreover, the ratings were initially created as a tool for brand security rather than a research instrument, which is why access as well as publication of the data is restricted. Finally, it remains unclear how inconsistencies in source selection, reliability ratings and other categories rated by NewsGuard would affect the results of a study. Therefore, we propose to provide a comprehensive description of this database over time and across countries by addressing the following aspects of the database:

- a) the distribution, construction, weighting and update process of the ratings,

- b) the selection and removal of sources, and
- c) the completeness and reliability of other information provided in the database, e.g., political orientation.

In addition, we reproduce published research relying on the NewsGuard list with different versions of the list to assess the degree to which results depend on a specific version of the list.

How is the sample constructed?

NewsGuard describes that they derive the source list from data services tracking 95% of all online activity. Notably, they add sources they deem influential, or that frequently change domains. In the database, the list automatically updates every hour. Given that we have access to the entire NewsGuard database, we constructed a sample by merging all lists with monthly granularity, covering March 2019 until March 2023. Over four years, the dataset has grown from 2,622 to 8,664 sources, and now covers 9 countries (United States, Italy, Great Britain, France, Germany, Canada, Australia, Austria, New Zealand) and five different languages (English, Italian, French, German, Spanish). Yet, the majority of the sources (76%) are in English and come from the United States. After analyzing global trends for the whole sample, we subset the non-U.S. data and compare across those countries.

How does it pertain to digital media?

When judging the consequences of misinformation at the scale of the information ecosystem (Allen et al., 2020), it feeds into a system shaped by growing partisan sorting and intergroup animosity (González-Bailón et al., 2023; Robertson et al., 2023). Misinformation is widely visible on social media (Lazer et al., 2018) and in the mainstream media (Tsfati et al., 2020), because it exploits the dynamics at play by fueling intergroup conflict and negative emotions. Consequently, to understand the spread of misinformation on digital platforms, it is imperative for computational social scientists to have access to reliable and valid data. By assessing the reliability of source-based approaches, we outline potential pitfalls for digital media research and offer recommendations for the application of the NewsGuard database in different countries.

References

- Allen, J., Howland, B., Mobius, M., Rothschild, D., and Watts, D. J. (2020). Evaluating the fake news problem at the scale of the information ecosystem. *Science Advances*, 6(14):eaay3539. Publisher: American Association for the Advancement of Science.
- González-Bailón, S., Lazer, D., Barberá, P., Zhang, M., Allcott, H., Brown, T., Crespo-Tenorio, A., Freelon, D., Gentzkow, M., Guess, A. M., Iyengar, S., Kim, Y. M., Malhotra, N., Moehler, D., Nyhan, B., Pan, J., Rivera, C. V., Settle, J., Thorson, E., Tromble, R., Wilkins, A., Wojcieszak, M., de Jonge, C. K., Franco, A., Mason, W., Stroud, N. J., and Tucker, J. A. (2023). Asymmetric ideological segregation in exposure to political news on Facebook. *Science*, 381(6656):392–398. Publisher: American Association for the Advancement of Science.
- Grinberg, N., Joseph, K., Friedland, L., Swire-Thompson, B., and Lazer, D. (2019). Fake news on Twitter during the 2016 U.S. presidential election. *Science*, 363(6425):374–378. Publisher: American Association for the Advancement of Science Section: Research Article.
- Guess, A. M., Nyhan, B., and Reifler, J. (2020). Exposure to untrustworthy websites in the 2016 US election. *Nature Human Behaviour*, 4(5):472–480. Number: 5 Publisher: Nature Publishing Group.
- Lasser, J., Aroyehun, S. T., Simchon, A., Carrella, F., Garcia, D., and Lewandowsky, S. (2022). Social media sharing by political elites: An asymmetric American exceptionalism. arXiv:2207.06313 [cs].
- Lazer, D., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., Nyhan, B., Pennycook, G., Rothschild, D., Schudson, M., Sloman, S. A., Sunstein, C. R., Thorson, E. A., Watts, D. J., and Zittrain, J. L. (2018). The science of fake news. *Science*, 359(6380):1094–1096.
- Lin, H., Lasser, J., Lewandowsky, S., Cole, R., Gully, A., Rand, D. G., and Pennycook, G. (2023). High level of correspondence across different news domain quality rating sets. *PNAS Nexus*, page pgad286.
- Robertson, R. E., Green, J., Ruck, D. J., Ognyanova, K., Wilson, C., and Lazer, D. (2023). Users choose to engage with more partisan news

than they are exposed to on Google Search. *Nature*, 618(7964):342–348.

Tsfati, Y., Boomgaarden, H. G., Strömbäck, J., Vliegenthart, R., Damstra, A., and Lindgren, E. (2020). Causes and consequences of mainstream media dissemination of fake news: literature review and synthesis. *Annals of the International Communication Association*, 44(2):157–173. Publisher: Routledge _eprint: <https://doi.org/10.1080/23808985.2020.1759443>.

Vosoughi, S., Roy, D., and Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380):1146–1151. Publisher: American Association for the Advancement of Science Section: Report.